

EE/CE 252/251

Signals and Systems

with

Dr. Naveed R. Butt

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Habib University

In the last lecture, we saw what the Z Transform is and what it can do for us.

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- Today, we go into details of the Z Transform, including:
 - **What values can z take?**
 - **What are some of the important properties of Z Transform?**
 - **How do we find Z Transform and its inverse?**
 - **How can we use Z Transform for system modeling, analysis, and design?**
 - **How can we solve LCCDEs using Z Transform?**

Last time we saw that z in the definition of Z Transform is a complex number and z^n is a complex exponential.

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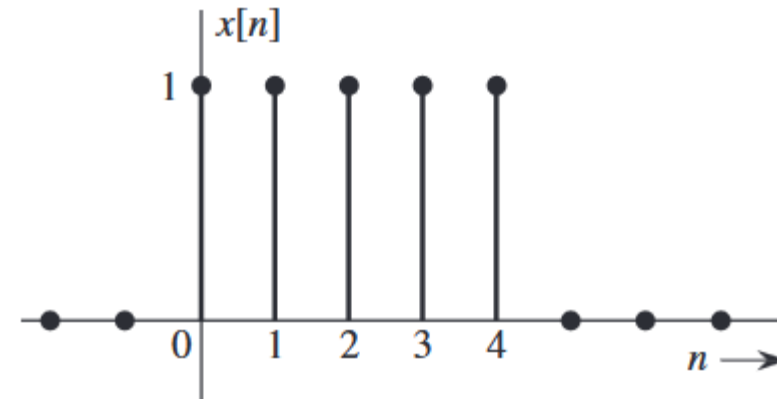
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$$\text{ROC} = \left\{ z : \left| \sum_{n=-\infty}^{\infty} x[n]z^{-n} \right| < \infty \right\}$$

Let's see a few examples of how the ROC may be different for different $x[n]$

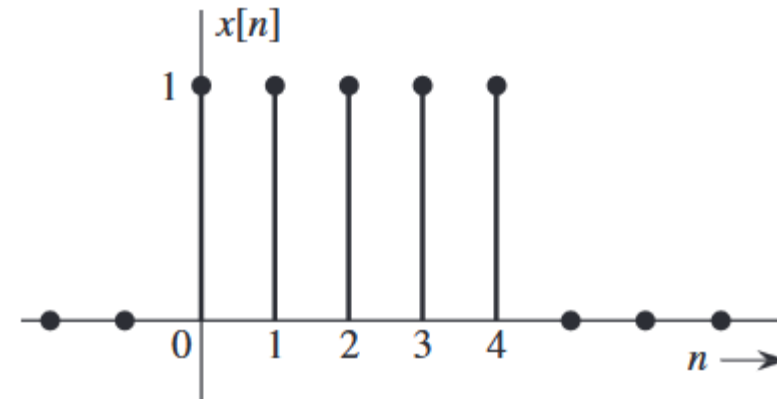
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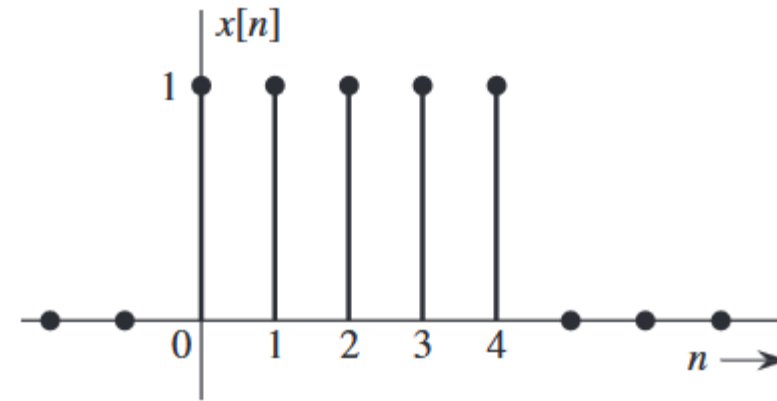
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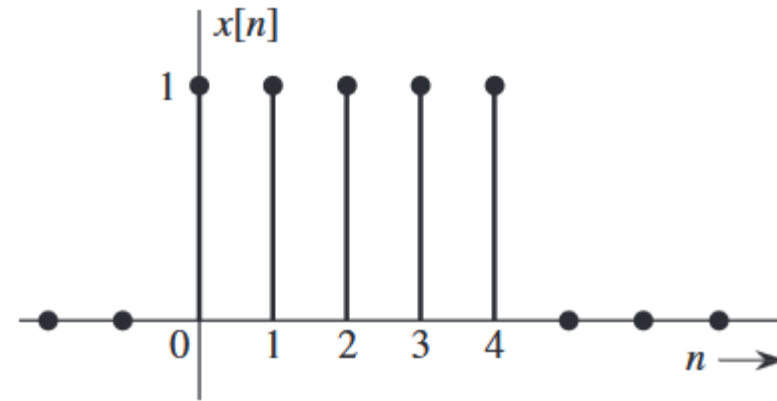
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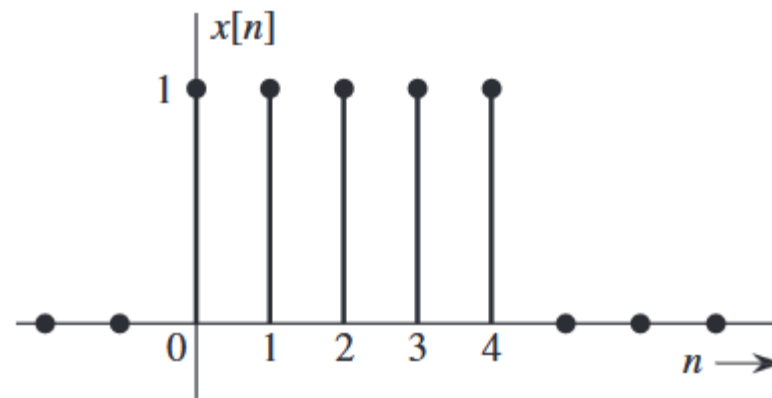


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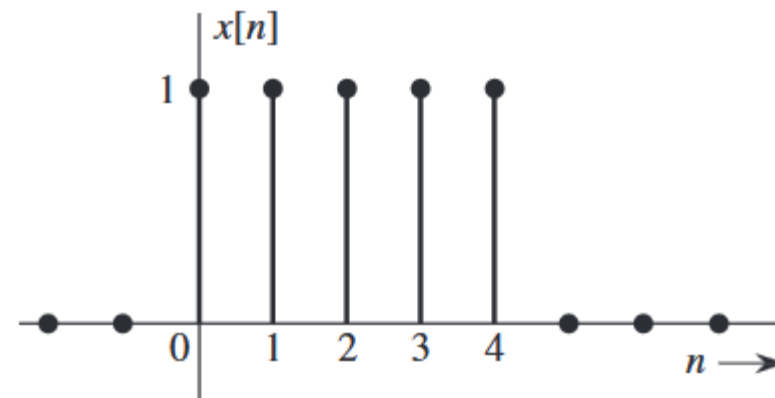
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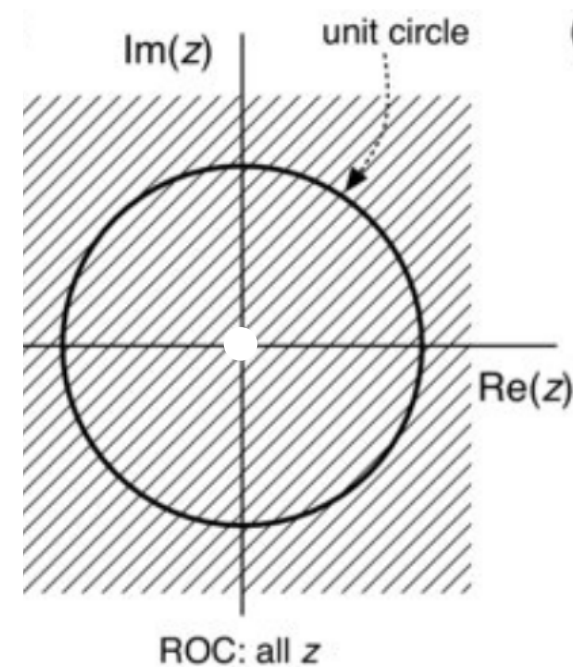


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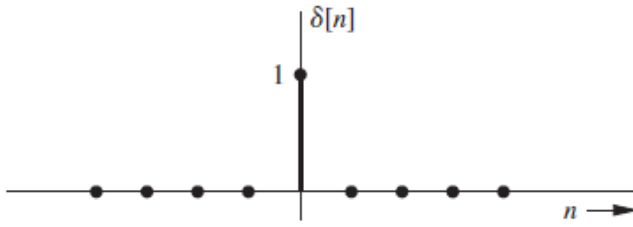
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ROC: $z \neq 0$
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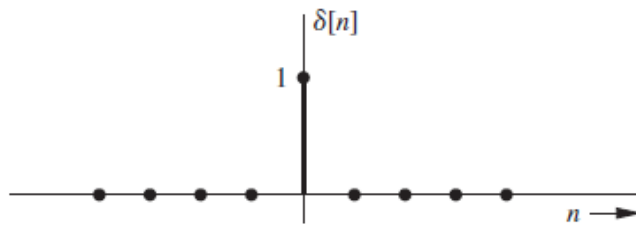


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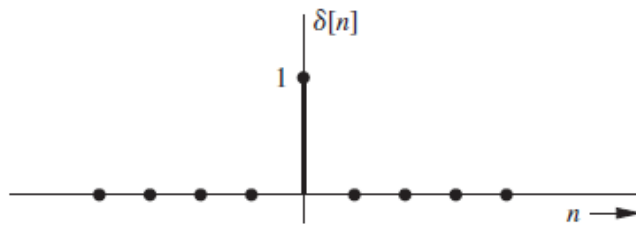
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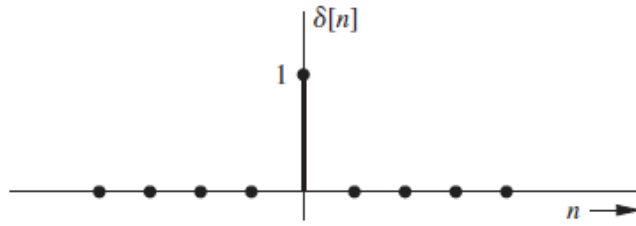


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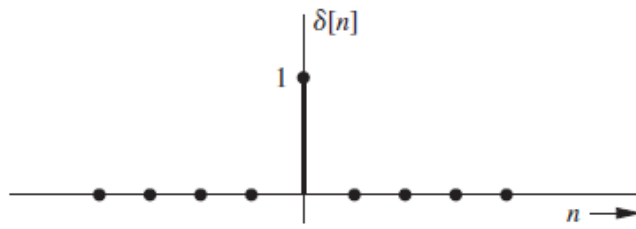
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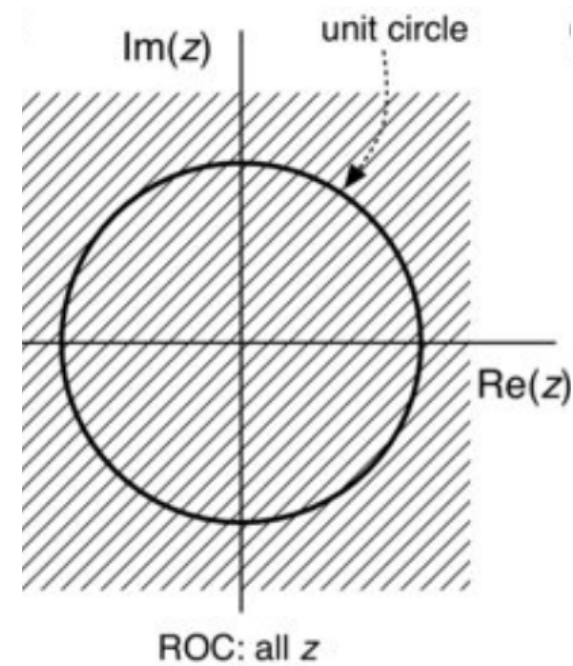
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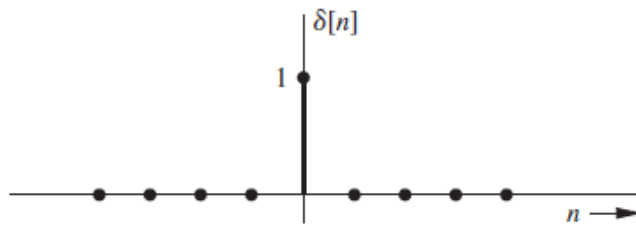
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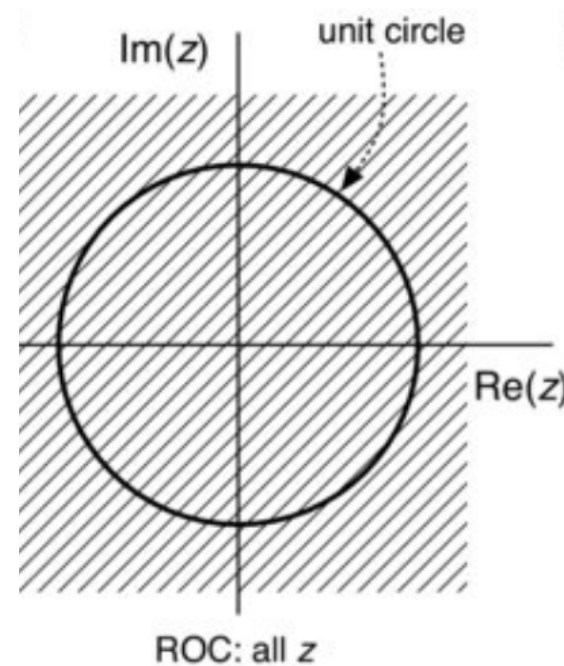
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Important Result:

$$\delta[n] \iff 1$$

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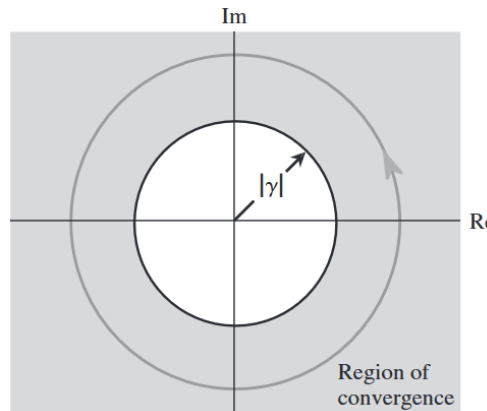
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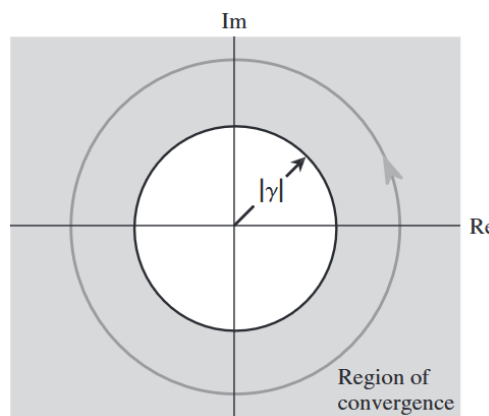
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Q. What are some of the important properties of Z Transform?

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$$\begin{aligned} \mathcal{Z}\{x[n - k]\} &= \sum_{n=0}^{\infty} x[n - k] z^{-n} \\ &= \sum_{j=-k}^{\infty} x[j] z^{-(j+k)} && j = n - k \\ &= \sum_{j=-k}^{\infty} x[j] z^{-j} z^{-k} \\ &= z^{-k} \sum_{j=-k}^{\infty} x[j] z^{-j} \\ &= z^{-k} \sum_{j=0}^{\infty} x[j] z^{-j} && x[\beta] = 0, \beta < 0 \\ &= z^{-k} X(z) \end{aligned}$$

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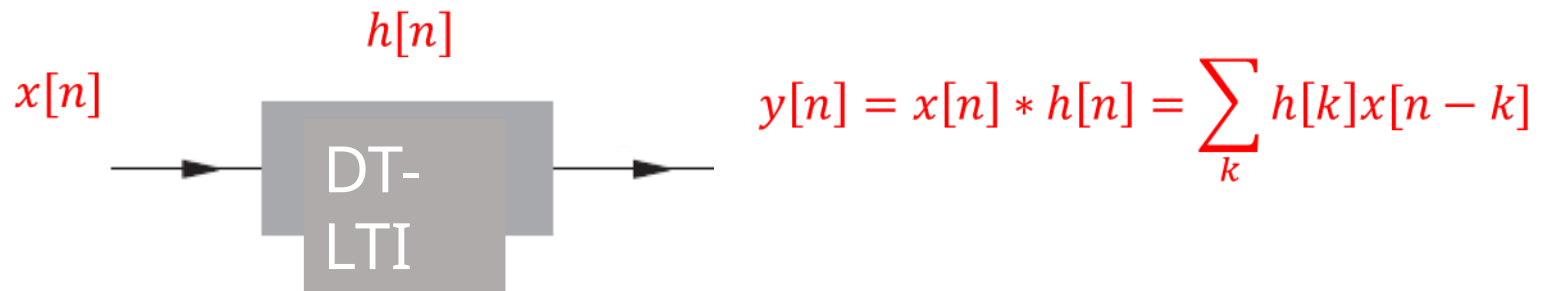
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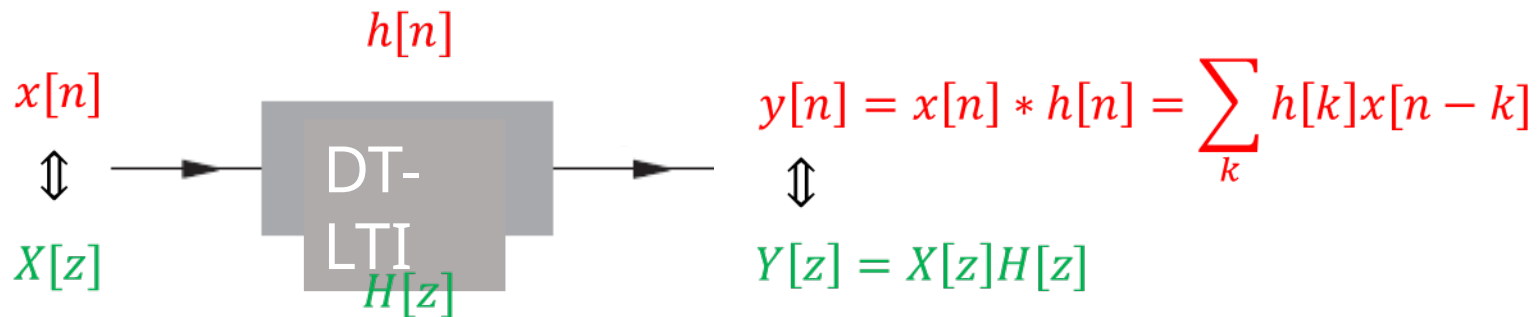


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becomes multiplication in z-domain

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This again shows that working in z-domain greatly simplifies finding output of a DT-LTI system.

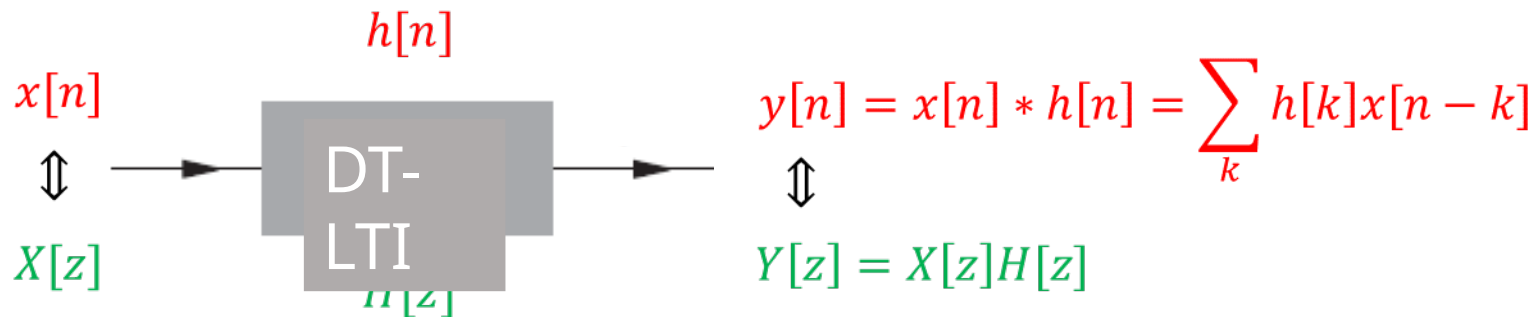


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Crux: since the basis/eigenfunctions are fixed, we only need to find the **product of "weights" and "eigenvalues"**.

5

Convolution in time-domain
becomes multiplication in z-domain

$$x_1[n] * x_2[n] \Leftrightarrow X_1[z]X_2[z]$$

Proof

$$\begin{aligned}\mathcal{Z}\{x_1[n] * x_2[n]\} &= \mathcal{Z}\left[\sum_{m=-\infty}^{\infty} x_1[m]x_2[n-m]\right] \\ &= \sum_{n=-\infty}^{\infty} z^{-n} \sum_{m=-\infty}^{\infty} x_1[m]x_2[n-m]\end{aligned}$$

$$\begin{aligned}\mathcal{Z}[x_1[n] * x_2[n]] &= \sum_{m=-\infty}^{\infty} x_1[m] \sum_{n=-\infty}^{\infty} x_2[n-m]z^{-n} \\ &= \sum_{m=-\infty}^{\infty} x_1[m] \sum_{r=-\infty}^{\infty} x_2[r]z^{-(r+m)} \\ &= \sum_{m=-\infty}^{\infty} x_1[m]z^{-m} \sum_{r=-\infty}^{\infty} x_2[r]z^{-r} \\ &= X_1[z]X_2[z]\end{aligned}$$

6

**Time-Reversal
Property**

if $x[n] \Leftrightarrow X(z)$
Then $x[-n] \Leftrightarrow X(1/z)$

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Proof

$$\mathcal{Z}\{x[-n]\} = \sum_{n=-\infty}^{\infty} x[-n]z^{-n}$$

$$\begin{aligned}\mathcal{Z}\{x[-n]\} &= \sum_{n=-\infty}^{\infty} x[n]z^n \\ &= \sum_{n=-\infty}^{\infty} x[n](1/z)^{-n} \\ &= X[1/z]\end{aligned}$$

7

Initial-Value
Theorem

$$x[0] = \lim_{z \rightarrow \infty} X[z]$$

8

Final-Value Theorem

$$\lim_{N \rightarrow \infty} x[N] = \lim_{z \rightarrow 1} (z - 1)X[z]$$

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If $x[n]$ is causal, we can find its initial value in frequency domain using:

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Final-Value Theorem

If $x[n]$ is causal, and $(z - 1)X[z]$ has no poles* outside the unit circle, then we can find its final (steady-state) value using:

$$\lim_{N \rightarrow \infty} x[N] = \lim_{z \rightarrow 1} (z - 1)X[z]$$

* We will define “poles” soon.

Q. How do we find the Z Transform?

Three common ways of finding the Z Transform

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Much like derivatives
and integrals!

A typical table of
Z Transform pairs.

$x[n]$	$X[z]$
$\delta[n - k]$	z^{-k}
$u[n]$	$\frac{z}{z - 1}$
$nu[n]$	$\frac{z}{(z - 1)^2}$
$n^2u[n]$	$\frac{z(z + 1)}{(z - 1)^3}$
$n^3u[n]$	$\frac{z(z^2 + 4z + 1)}{(z - 1)^4}$
$\gamma^n u[n]$	$\frac{z}{z - \gamma}$
$\gamma^{n-1} u[n - 1]$	$\frac{1}{z - \gamma}$
$n\gamma^n u[n]$	$\frac{\gamma z}{(z - \gamma)^2}$
$n^2\gamma^n u[n]$	$\frac{\gamma z(z + \gamma)}{(z - \gamma)^3}$
$\frac{n(n - 1)(n - 2) \cdots (n - m + 1)}{\gamma^m m!} \gamma^n u[n]$	$\frac{z}{(z - \gamma)^{m+1}}$
$ \gamma ^n \cos \beta n u[n]$	$\frac{z(z - \gamma \cos \beta)}{z^2 - (2 \gamma \cos \beta)z + \gamma ^2}$
$ \gamma ^n \sin \beta n u[n]$	$\frac{z \gamma \sin \beta}{z^2 - (2 \gamma \cos \beta)z + \gamma ^2}$

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Find the Z Tx of $x[n] = (5 + 5^n)u[n]$

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Using Linearity and the transform pairs (from the table), we get

$$u[n] \Leftrightarrow \frac{z}{z-1}$$

$$\gamma^n u[n] \Leftrightarrow \frac{z}{z-\gamma}$$

$$X[z] = \frac{5z}{z-1} + \frac{z}{z-5}$$

Example 2

Find the Z Tx of $x[n] = 5^{n+1}u[n]$

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In order to make use of known pairs, we can rewrite $x[n]$ as

$$x[n] = 5(5^n u[n])$$

Using homogeneity (part of linearity) and the transform pair (from the table), we get

$$\gamma^n u[n] \Leftrightarrow \frac{z}{z - \gamma}$$

$$X[z] = \frac{5z}{z - 5}$$

Example 3

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We can use shifting
property

$$x[n-k] \Leftrightarrow z^{-k}X[z] + z^{-k} \sum_{m=1}^k x[-m]z^m$$

And transform
pair

$$\gamma^n u[n] \Leftrightarrow \frac{z}{z-\gamma}$$

To get

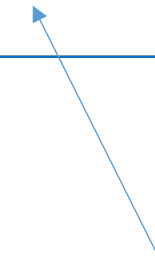
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And transform pair

$$\gamma^n u[n] \Leftrightarrow \frac{z}{z - \gamma}$$

Note that these terms are all zeros since $u[-1] = u[-2] = u[-3] = u[-4] = 0$

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Methods used in practice...

- Numerically (MATLAB, Python etc.)
- Using Partial Fraction Expansion
 - Along with tables and properties
- Using Long Division
 - Solution not closed form

Partial Fraction Expansion (PFE)

- Often covered in college math
- See textbook Chapter B (Background) for a quick revision

Inverse Z Transform Using PFE Method

- Step 1 : Apply PFE to $X[z]/z$
- Step 2: Multiply by z
- Step 3: Use transform-pair table

Inverse Z Transform Using PFE Method

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Experience shows that $X[z]/z$ leads to a better time-form (e.g., $u[n]$ and not $u[n - 1]$)

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To remove the effect of division by z

Mostly this pair is needed.

$$\gamma^n u[n] \Leftrightarrow \frac{z}{z - \gamma}$$

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Find inverse Z Tx of $\frac{8z - 19}{(z - 2)(z - 3)}$

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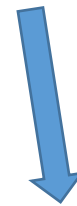
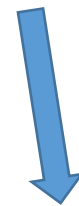
$$\frac{X[z]}{z} = \frac{8z - 19}{z(z - 2)(z - 3)} = \frac{(-19/6)}{z} + \frac{(3/2)}{z - 2} + \frac{(5/3)}{z - 3}$$

Step 2 – Multiplying by z

$$X[z] = -\frac{19}{6} + \frac{3}{2} \left(\frac{z}{z-2} \right) + \frac{5}{3} \left(\frac{z}{z-3} \right)$$

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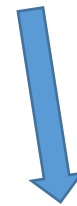
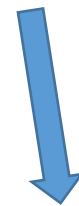


Step 3 – Using transform pairs to get $x[n]$

$$x[n] = -\frac{19}{6} \delta[n] + \left[\frac{3}{2} (2)^n + \frac{5}{3} (3)^n \right] u[n]$$

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HW: Go through examples of repeated real roots, and complex roots in textbook.

Q. How can we use Z Transform for system modeling, analysis, and design?

Rational Transfer Function

General LTI System **Model in Frequency Domain**

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Recall that the general model for DT-LTI systems we developed in time-domain (allowing for initial conditions) led to the LCCDE

$$y[n] = -a_1y[n-1] - a_2y[n-2] - \dots - a_Ny[n-N] \\ + b_0x[n] + b_1x[n-1] + \dots + b_Nx[n-N]$$

Linear Constant-Coefficient Difference Equation (LCCDE)

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Linear Constant-Coefficient Difference Equation (LCCDE)

Recall that this model also includes impulse response as a special case (zero initial conditions, and $x[n] = \delta[n]$).

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- We will now show that a general DT-LTI system can be modelled more *efficiently* in frequency domain.

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Can you take Z Transform of the general time-domain model above? (for convenience we can consider zero initial conditions and causal input).

$$y[n] = -a_1y[n-1] - a_2y[n-2] - \dots - a_Ny[n-N] \\ + b_0x[n] + b_1x[n-1] + \dots + b_Nx[n-N]$$



$$Y[z] = -a_1z^{-1}Y[z] - a_2z^{-2}Y[z] - \dots - a_Nz^{-N}Y[z] \\ + b_0X[z] + b_1z^{-1}X[z] + \dots + b_Nz^{-N}X[z]$$

Notice the excellent role
of z^{-1} in converting time-
delay to polynomials in z

$$x[n-k] \Leftrightarrow z^{-k}X[z]$$

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Taking $Y[z]$ terms to the left and multiplying out by z^N gives

$$(z^N + a_1z^{N-1} + \dots + a_{N-1}z + a_N)Y[z] \\ = (b_0z^N + b_1z^{N-1} + \dots + b_{N-1}z + b_N)X[z]$$

*We can thus write the output
as*

$$\begin{aligned} Y[z] &= \left(\frac{b_0 z^N + b_1 z^{N-1} + \dots + b_{N-1} z + b_N}{z^N + a_1 z^{N-1} + \dots + a_{N-1} z + a_N} \right) X[z] \\ &= \frac{P[z]}{Q[z]} X[z] \end{aligned}$$

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$$Y[z] = X[z]H[z].$$

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- It provides a much simpler system model (difference equation replaced by polynomials in z).

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- We call $H[z]$ the **Transfer Function**.
- It provides a much simpler system model (difference equation replaced by polynomials in z).
- Derived using zero initial conditions (i.e., zero-state case).
- It is not difficult to include initial conditions when required.

Let's get to know $H[z]$ a bit better...

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$H[z]$ as a **ratio of output to input**

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$H[z]$ in **factorized** form

$$H[z] = b_0 \frac{(z - z_1)(z - z_2) \cdots (z - z_N)}{(z - \gamma_1)(z - \gamma_2) \cdots (z - \gamma_N)}$$

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Roots ($z_1, \gamma_1, \dots$) of the numerator and the denominator of $H[z]$ are among the **most important design & analysis parameters** used in modern engineering.

Let's look at these in detail.

A Game of **Poles** and **Zeros**

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 - These are the roots of the denominator of $H[z]$

Example

$$H[z] = \frac{7z - 6}{z^2 - z - 6} \quad \longrightarrow \quad H[z] = \frac{1}{7} \frac{z - 6/7}{(z + 2)(z - 3)}$$

- Zeros: $z = \frac{6}{7}$
- Poles: $z = -2, z = 3$

Pole-Zero Plot

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 - Poles marked as $\times$
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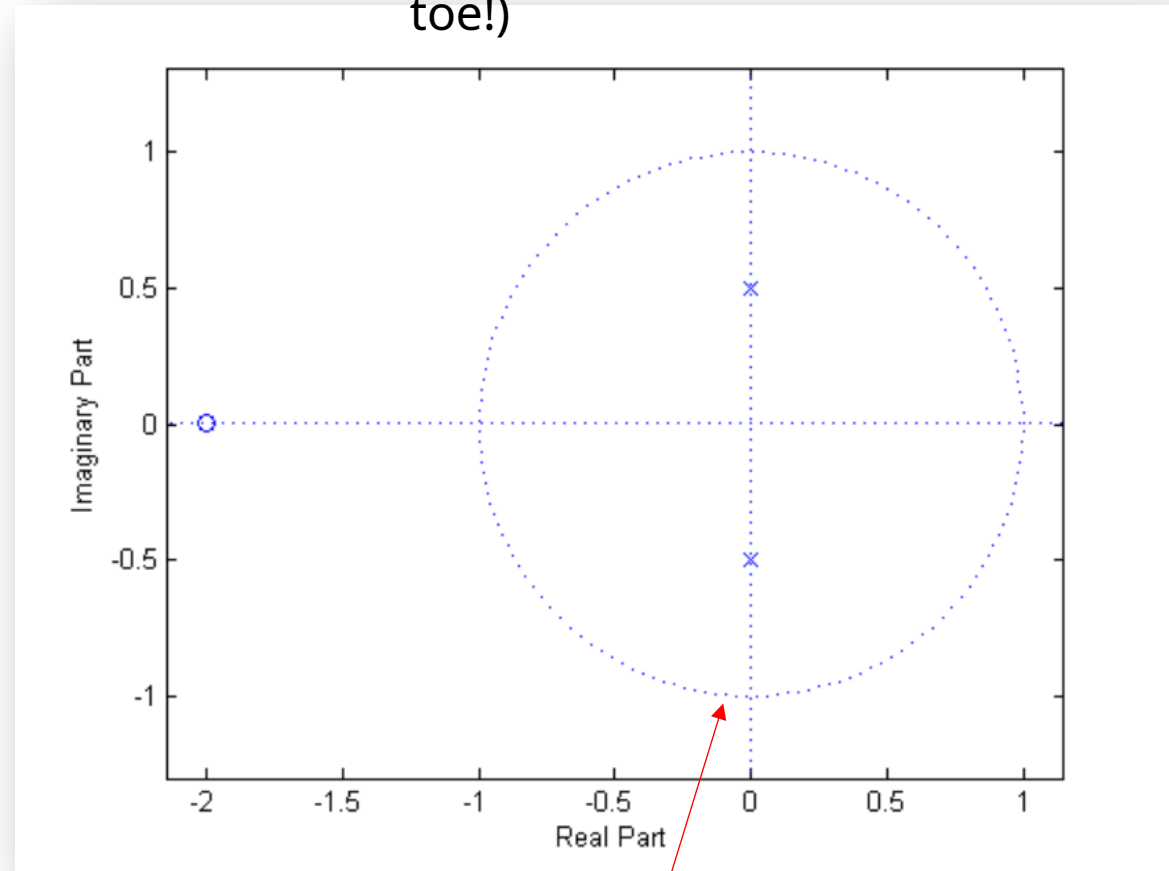
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(kind of like, an engineer's tic-tac-toe!)



Unit circle marked for reference

System Analysis Using Poles and Zeros of $H[z]$

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BIBO Stability

- A DT-LTI system is BIBO stable if all the poles of $H[z]$ are inside the unit circle.
 - i.e., for every pole z , we have $|z| < 1$

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- And the fact that BIBO stability requires $\sum |h[n]| < \infty$

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- Notice that a pole (γ) in z becomes base of a geometric series in time.
- Clearly corresponding geometric series will be convergent $|\gamma| < 1$,
- This is equivalent to saying that pole is inside the unit circle (on complex plane).

Example 1

$$H[z] = \frac{z}{z-2} \Leftrightarrow 2^n u[n] = h[n]$$

Pole
outside
unit circle



Divergent
series



**System
BIBO
unstable!**

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$$H[z] = \frac{z}{z-0.5} \Leftrightarrow (0.5)^n u[n] = h[n]$$

Pole outside
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circle

Converge
nt series

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Example 3

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Pole outside
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Example 3

$$H[z] = \frac{8z - 19}{(z - 0.2)(z - 3)} = \frac{az}{z - 0.2} + \frac{bz}{z - 3}$$

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Convergent

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Divergent

Thus, even if one pole is on or outside the unit circle, it will lead to a divergent series $h[n]$!

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Invertibility

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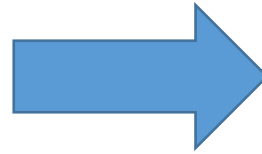
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- Thus the inverse, $H^{-1}[z]$, will be stable only if all the zeros of $H[z]$ are inside the unit circle.

Since we are interested in system inverses that are stable, we generally say a system is invertible if it is stable invertible.

Example

$$H[z] = \frac{(z - 0.5)(z + 3)}{(z + 0.2)}$$

Not stable-invertible, since one of the zeros ($z = -3$) is outside the unit circle.



$$G[z] = H^{-1}[z] = \frac{(z + 0.2)}{(z - 0.5)(z + 3)}$$

Easy to see that the inverse is an unstable system with a pole ($z = -3$) outside the unit circle.

Frequency Response Using Poles and Zeros of $H[z]$

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- Covered in great detail in courses such as DSP.
- Here we will just give a very basic idea.

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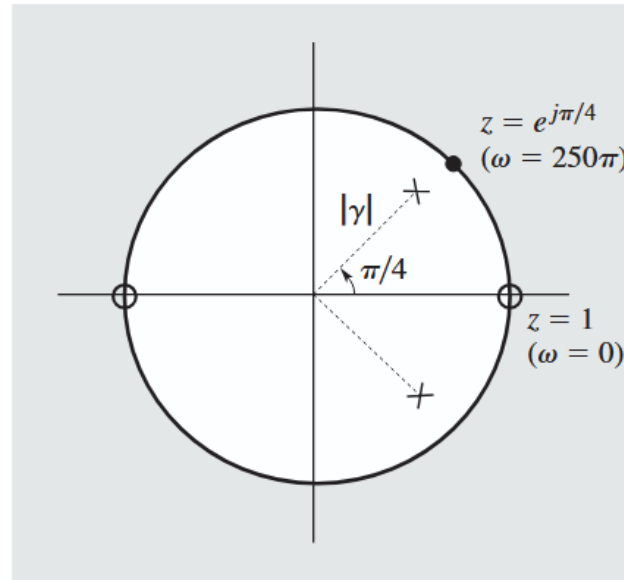
- **Zeros control dips:** giving locations of frequencies that may be attenuated and how strong the attenuation will be.
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Frequency Response Using Poles and Zeros of $H[z]$

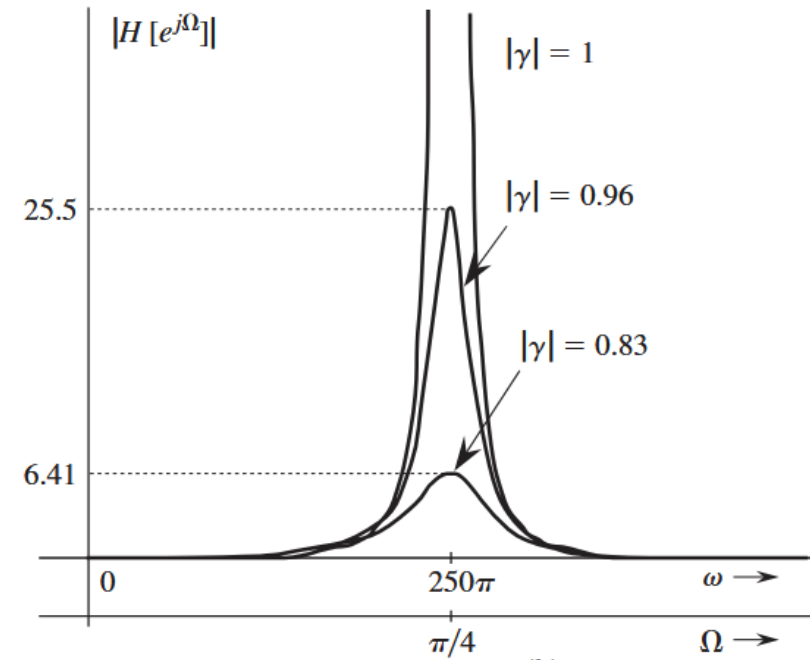
Some rules of thumb (without going into the math)

- **Zeros control dips:** giving locations of frequencies that may be attenuated and how strong the attenuation will be.
- **Poles control peaks:** giving locations of frequencies that may be amplified and how strong the amplification will be.
- **Poles and zeros work together** to create a complete net system response to different frequencies. They can reinforce or cancel each other's effect.

Example 1

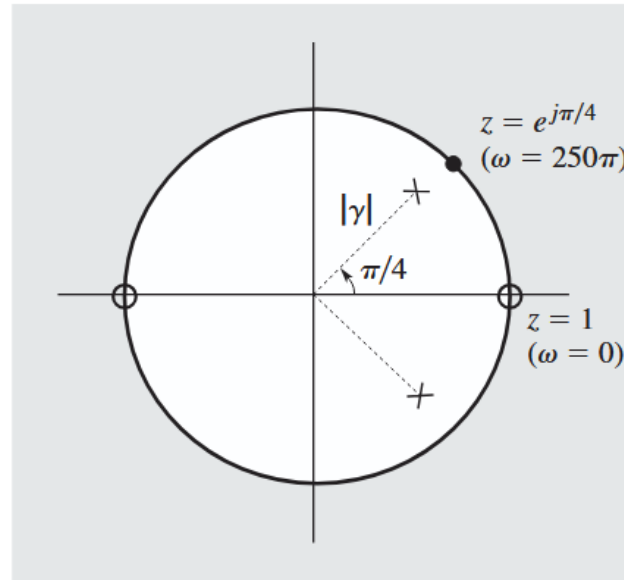


(a)

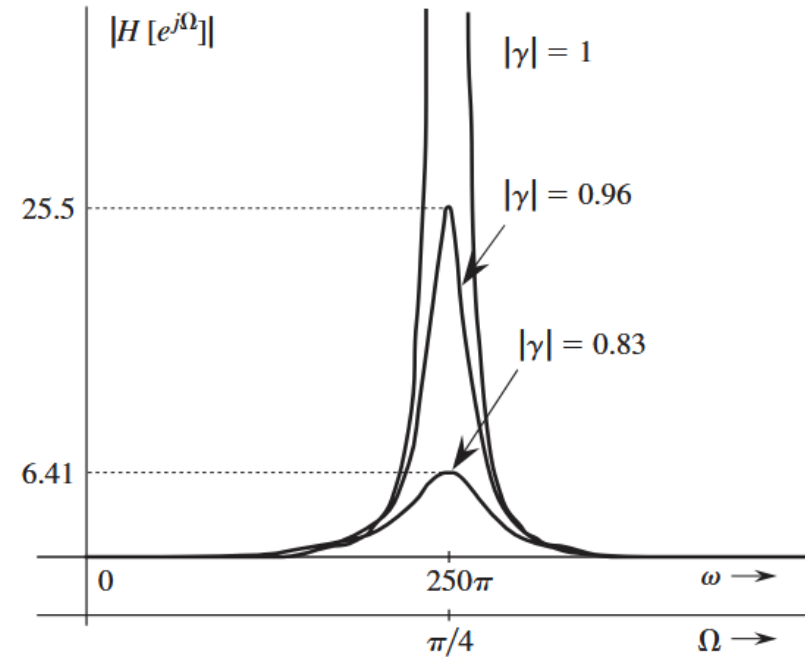


(b)

Example 1



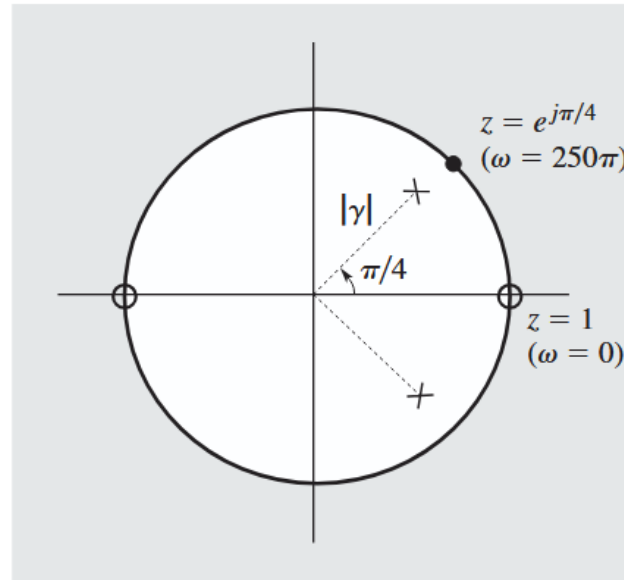
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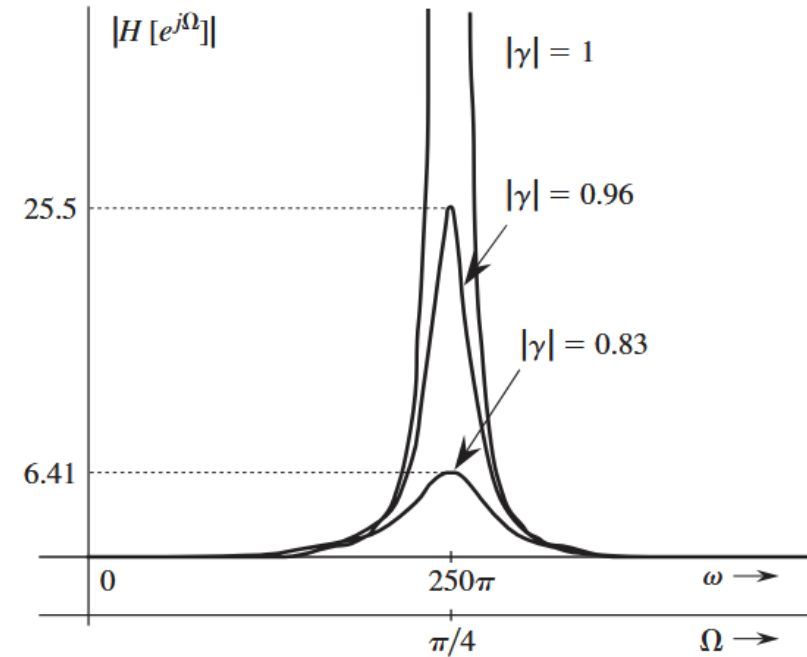
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- This system says, I will amplify sinusoids of frequency $\pm \pi/4$ radians if they are in the input.
- The amplification amount will increase as the poles move away from origin towards the unit circle.
- If they reach or cross the unit circle, system will become unstable.

Example 1



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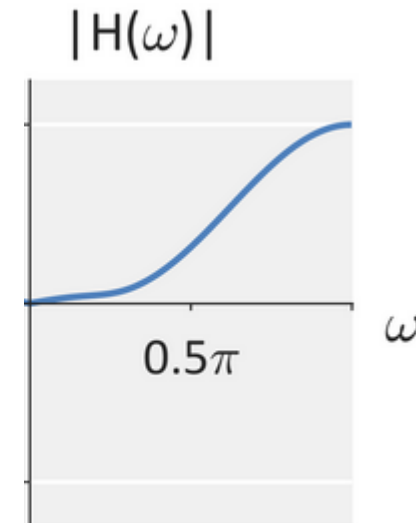
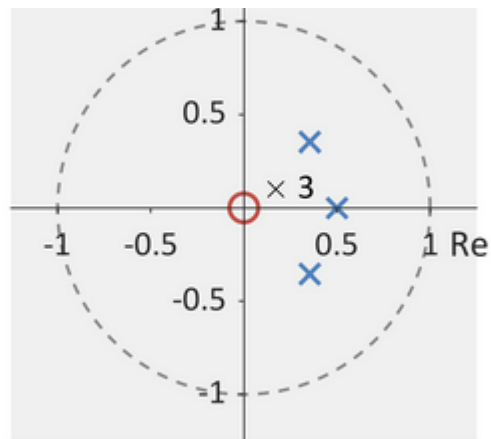
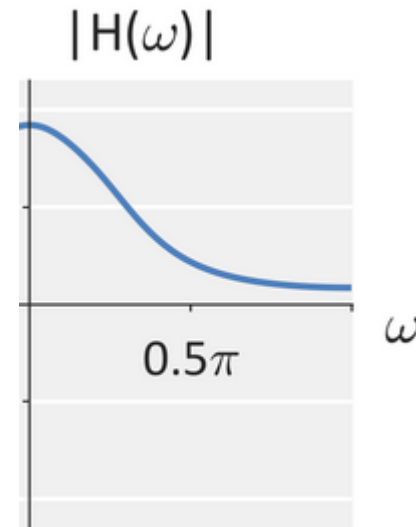
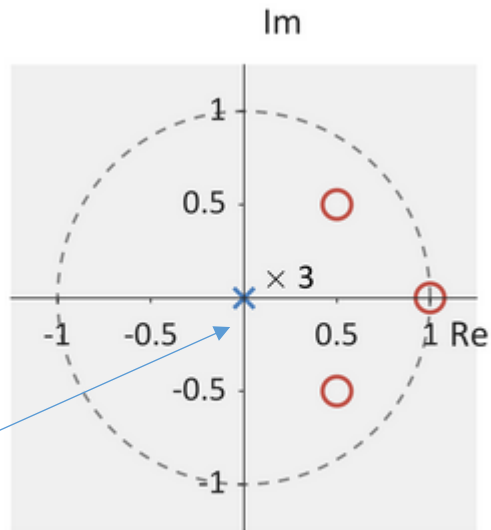
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- The amplification amount will increase as the poles move away from origin towards the unit circle.
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- Zeros at 0 and π radians say that frequencies at and near these will be attenuated.
- Together, the poles and zeros form this response.

Example 2

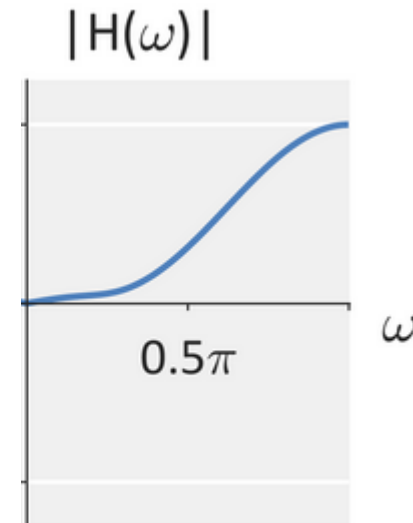
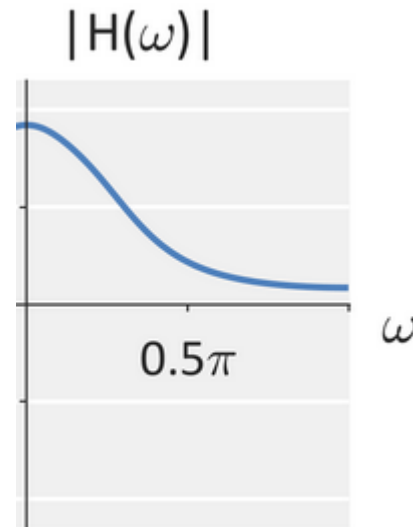
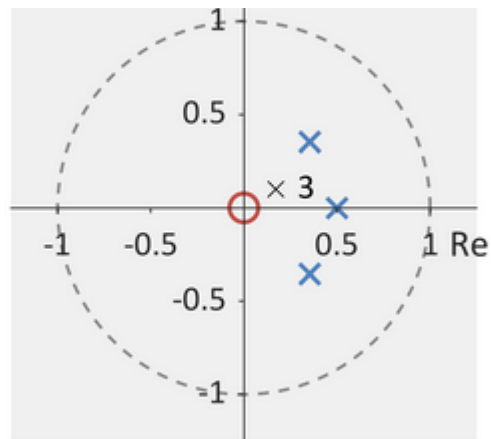
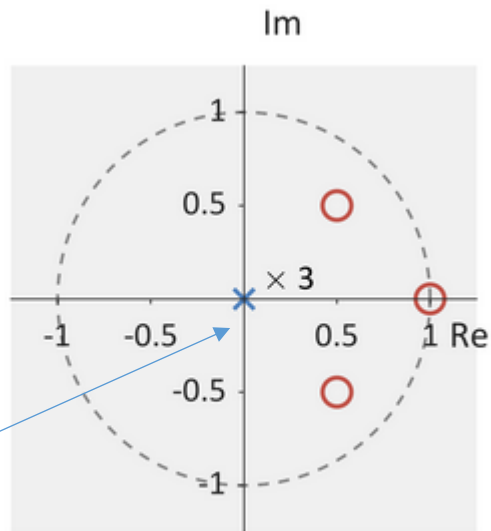
Can you guess which pole-zero plot corresponds to which frequency response?



$\times 3$ indicated that there are three (repeated) poles at this point.

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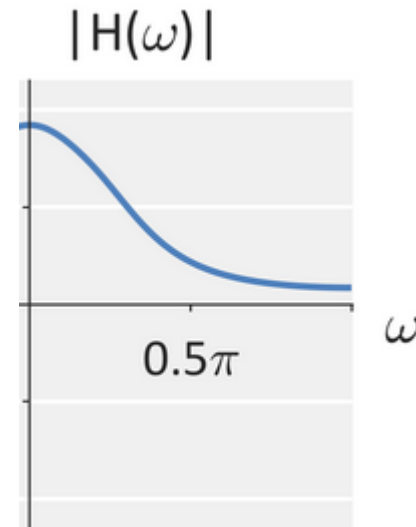
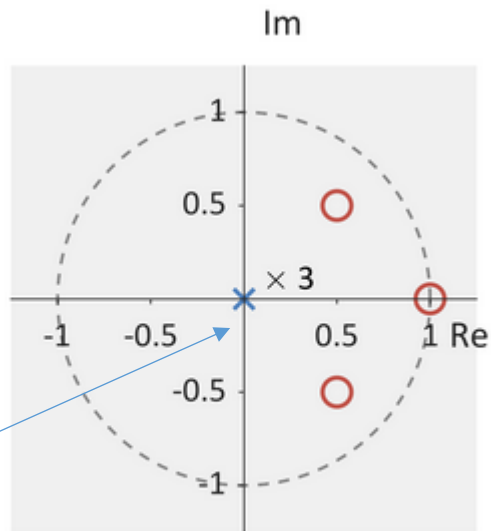
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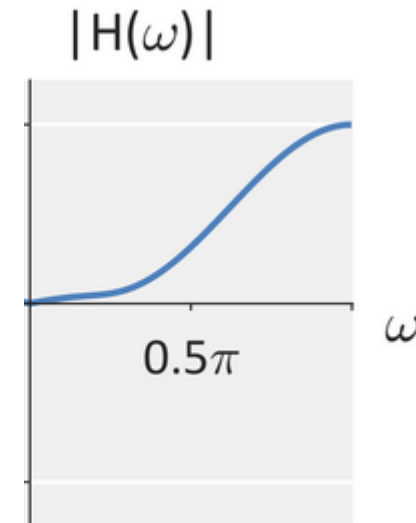
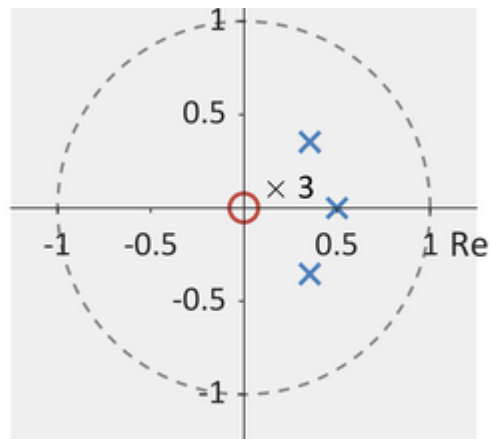
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Low-Pass Filter

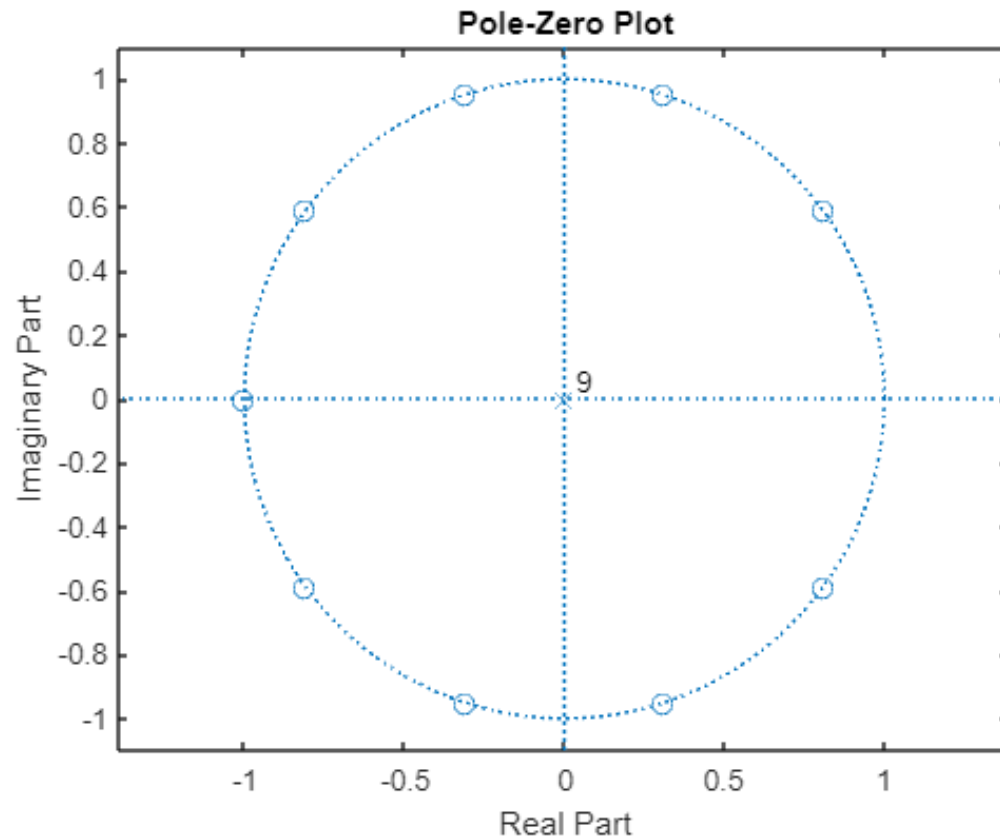


High-Pass Filter

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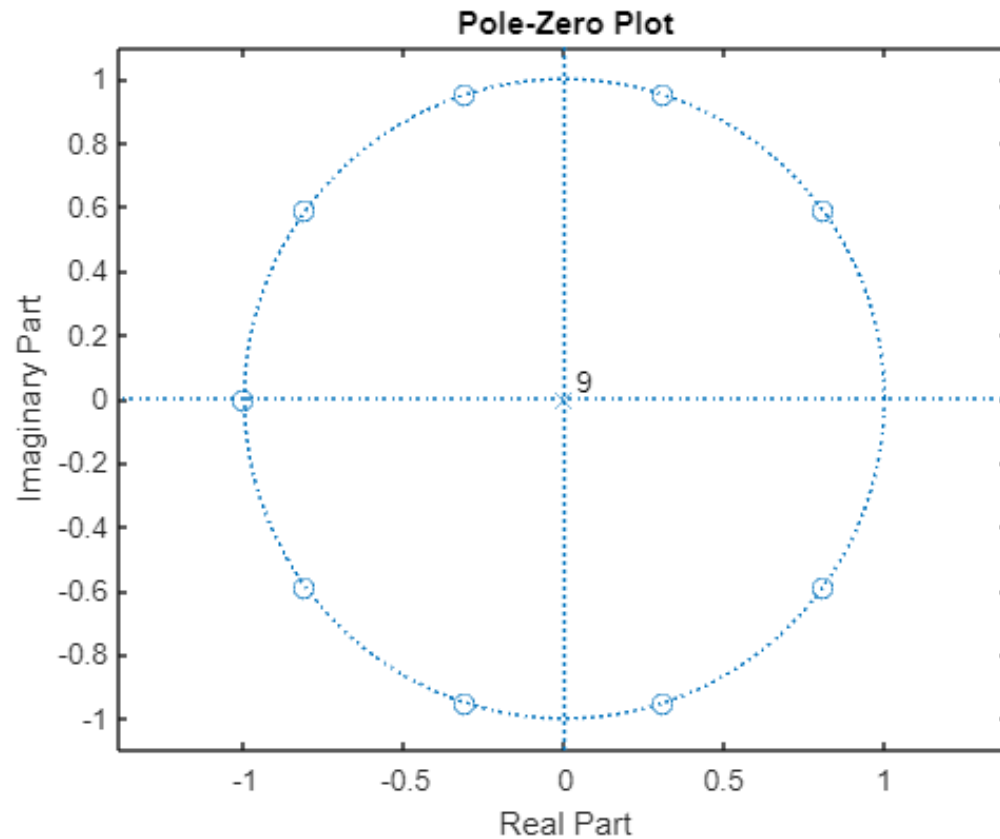
Example 3

Can you draw a rough sketch of the frequency response of this system?

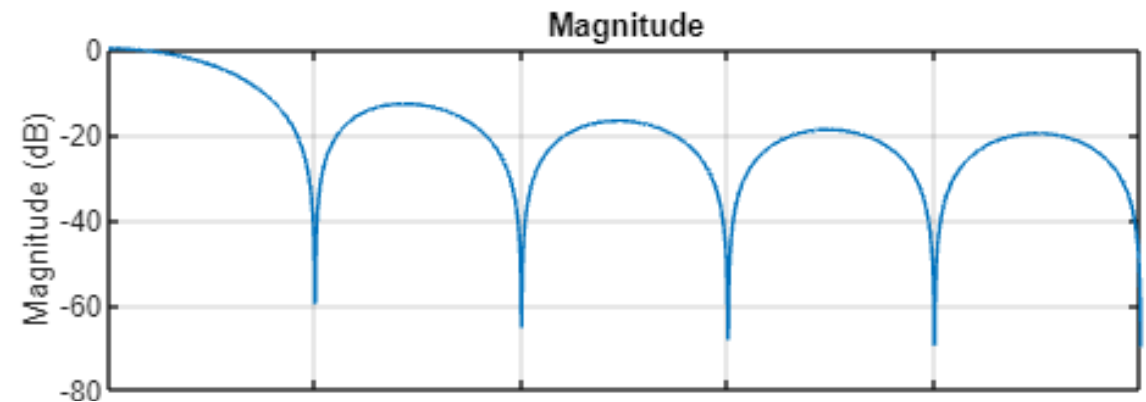


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Comb
Filter



System Design Using Frequency Domain

Two examples:

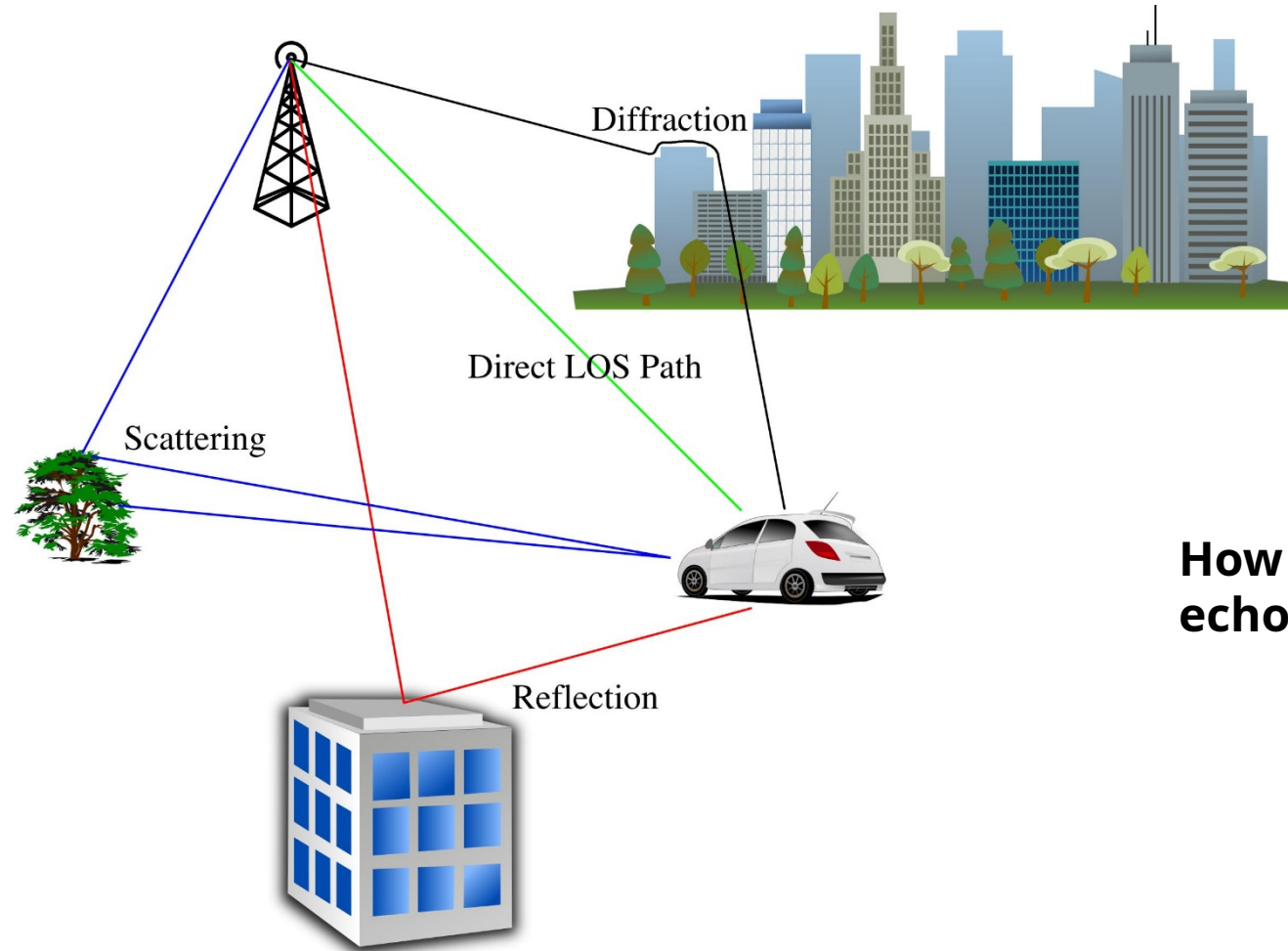
- There are situations where we would like to cancel the effect of a system.
 - This is relatively easy in frequency domain as we can make use of $H^{-1}[z]$ (provide system is invertible).
- There are situation where we have control over the parameters (physical or digital) of a system.
 - By tuning these, we can vary $H[z]$ (and its poles and zeros). We can thus choose tuning parameter such that poles and zeros fall in the right place.

**Example
1**

Using Z Transform (Frequency Domain)
to Analyze and “Fix” the System

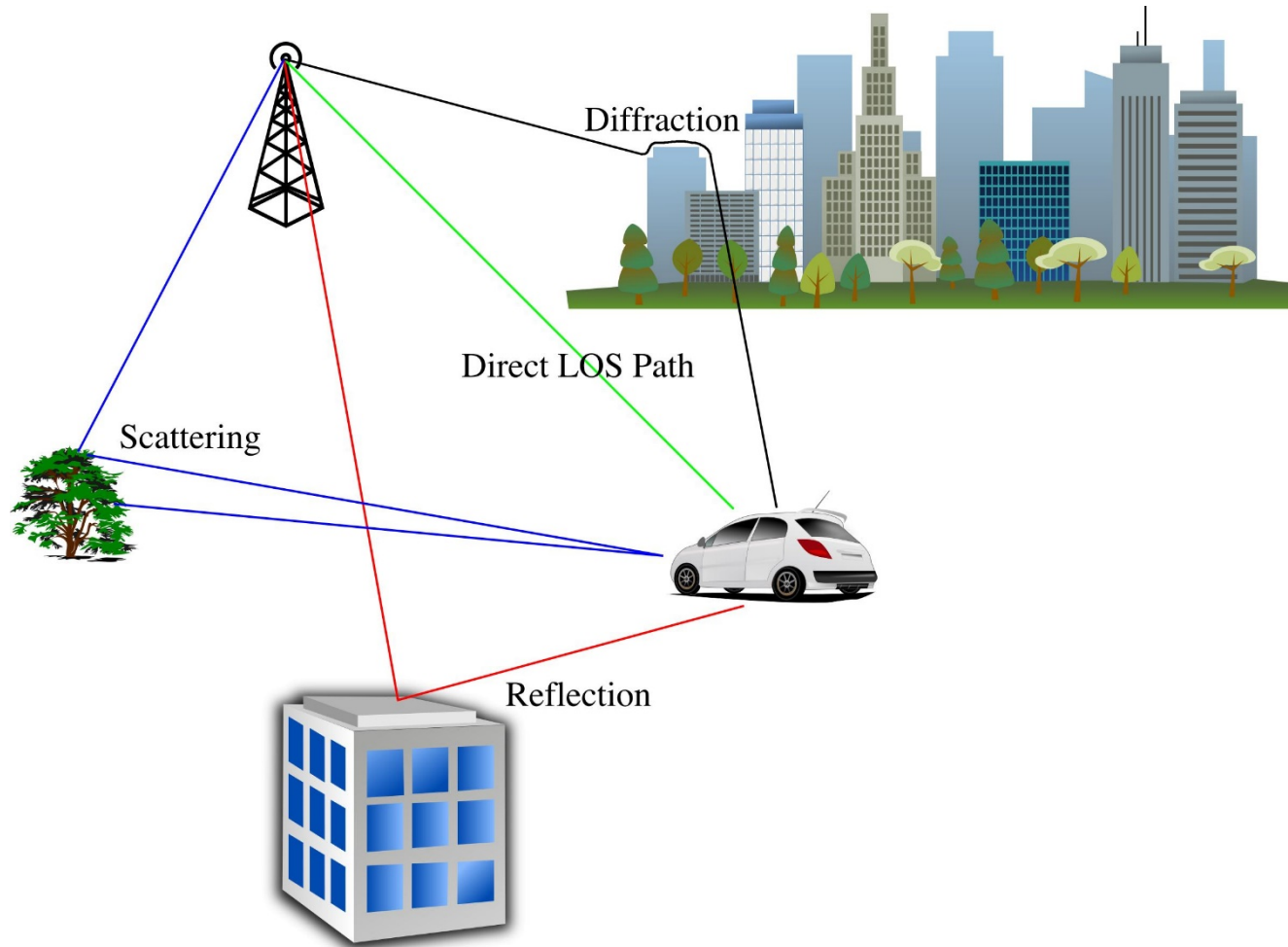
Multipath Signals in Wireless Communications

Design
Example



How do we model the signal (plus echoes) and remove the echoes?

Discrete-Time System Model

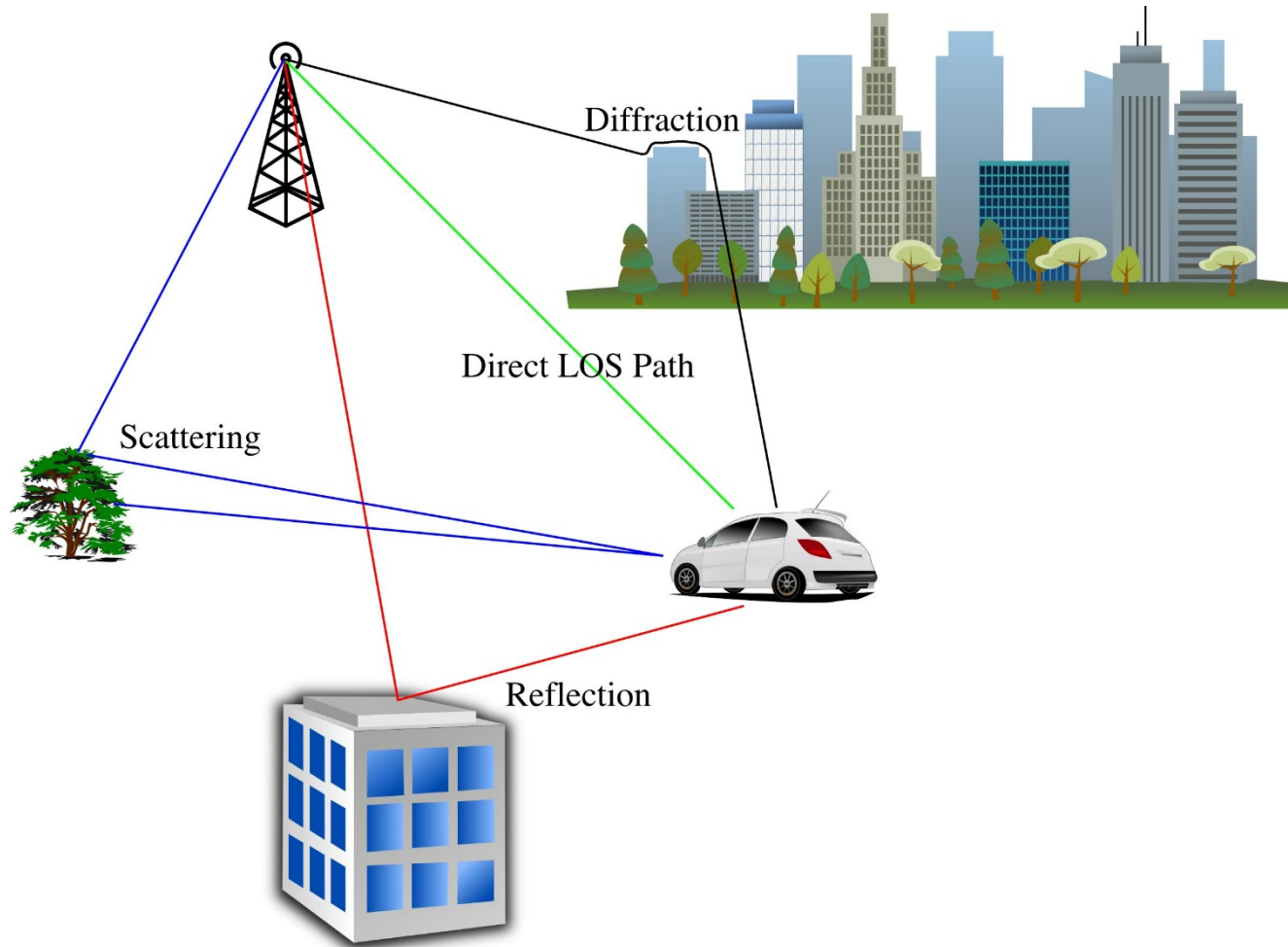


Perhaps only the **green** and **red** components are strong (and the rest are very weak)? If so, our model may look like:

$$y[n] = x[n] - \alpha x[n - N_0]$$

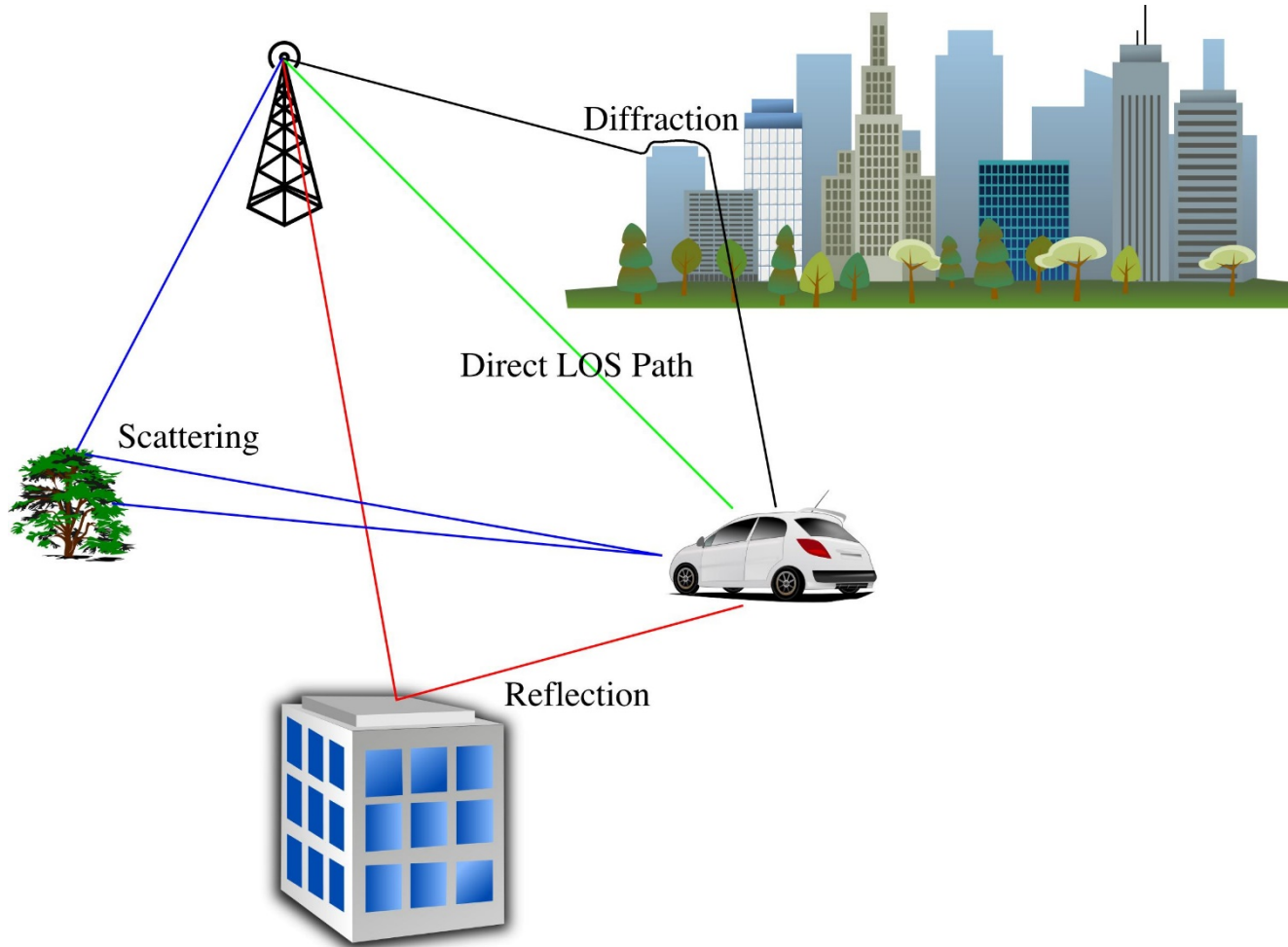
$$\alpha = 0.8, N_0 = 11$$

System in Frequency Domain



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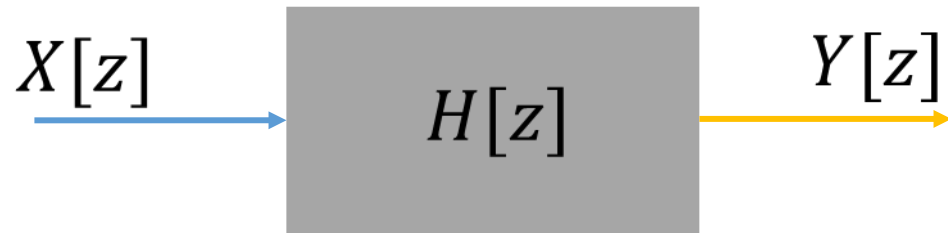
$$y[n] = x[n] - \alpha x[n - N_0]$$

$$Y[z] = X[z] - 0.8Z^{-11}X[z]$$

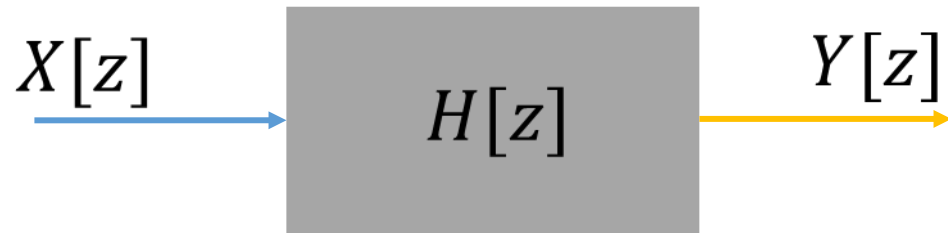
$$Y[z] = (1 - 0.8Z^{-11}) X[z]$$

“System” Transfer Function $H[z]$

Analyzing What the System Does

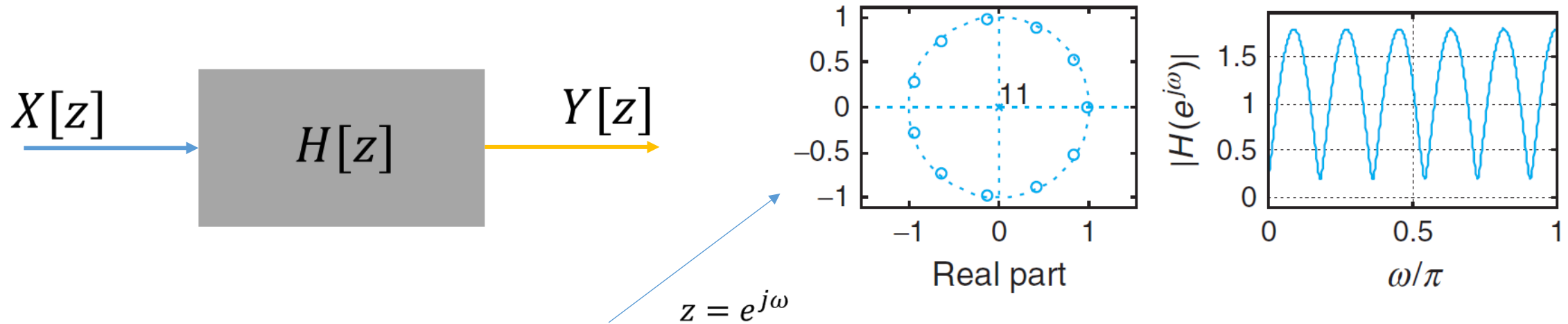


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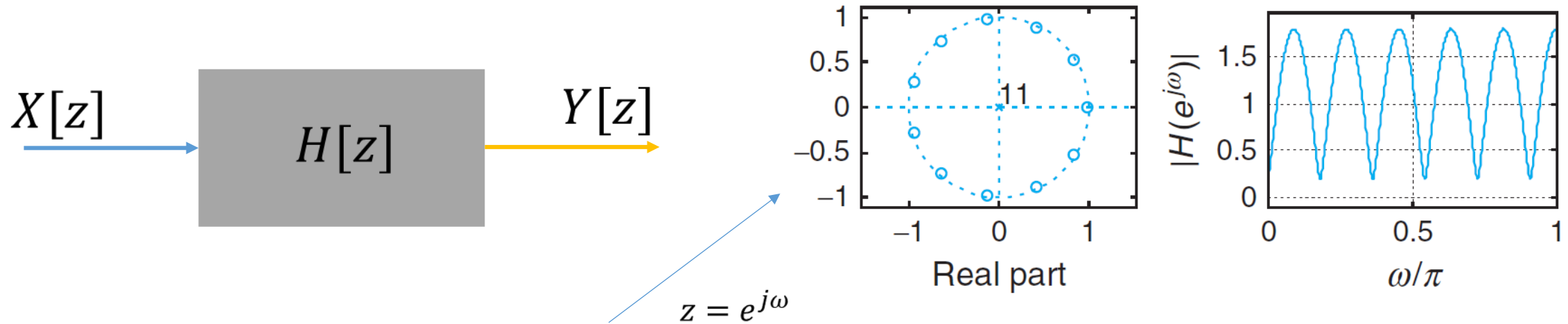
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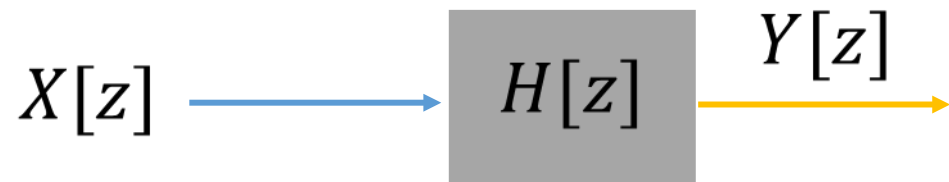
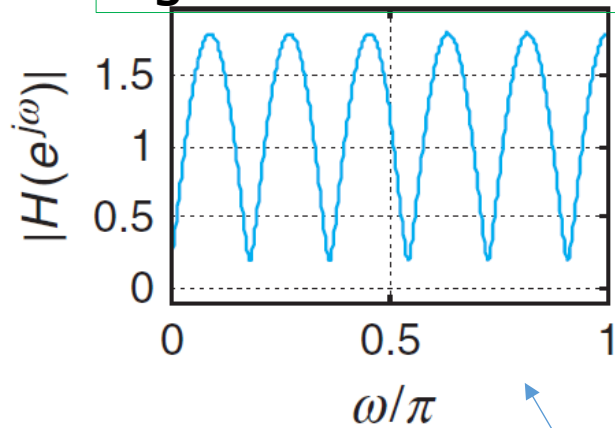
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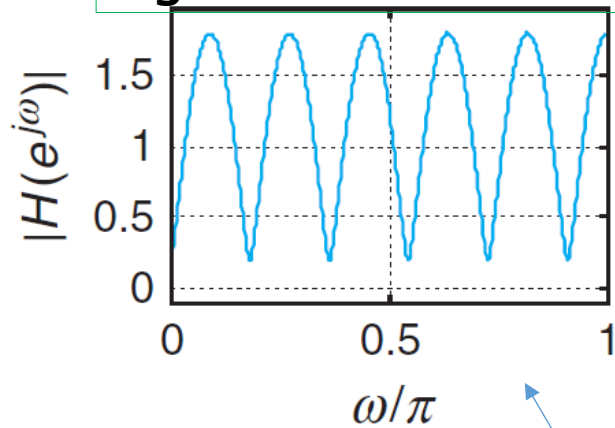
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- The environment behaves as a “comb” filter. **How may we remove this effect to get original signal back?**
- Note that $H[z]$ is invertible (all zeros inside the unit circle), so there is hope.

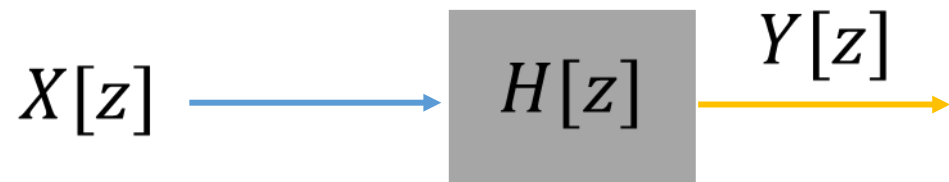
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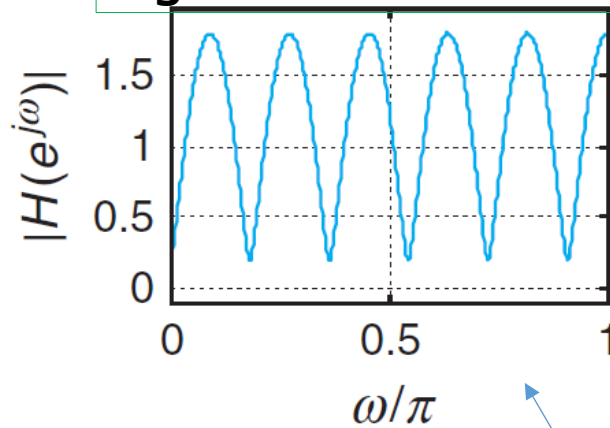
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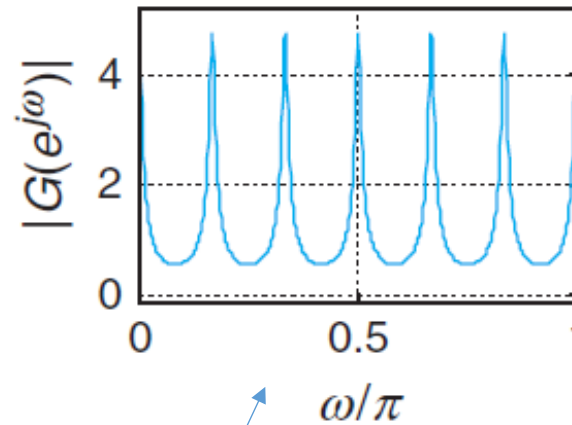
One way is to pass the received signal through a filter that cancels the frequency effect of the system

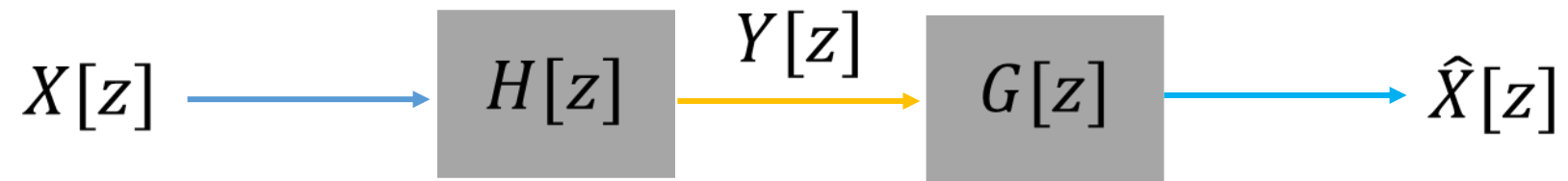


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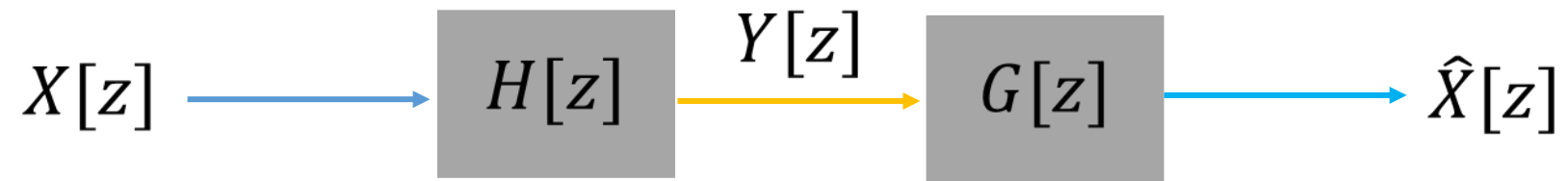


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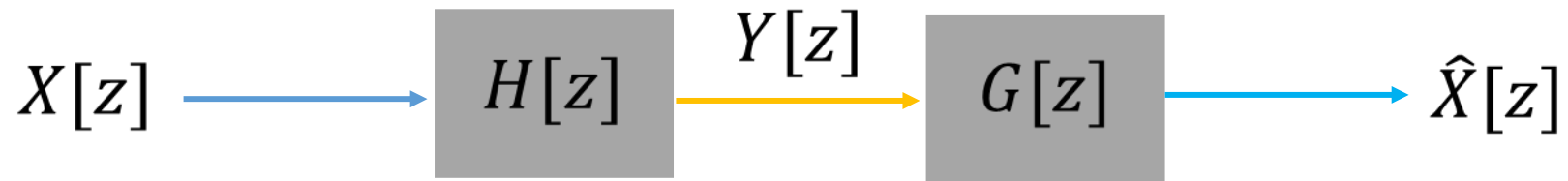




To get $\hat{\mathbf{X}}[\mathbf{z}] = \mathbf{X}[\mathbf{z}]$ we must have $\mathbf{H}[\mathbf{z}]\mathbf{G}[\mathbf{z}] = ?$



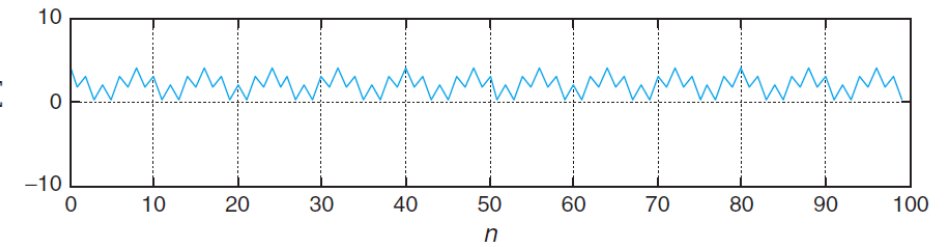
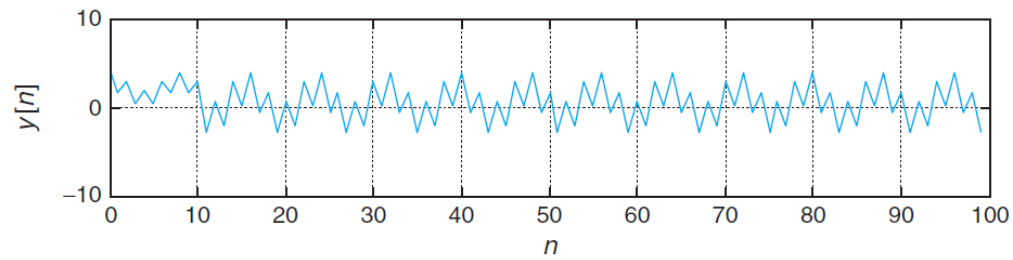
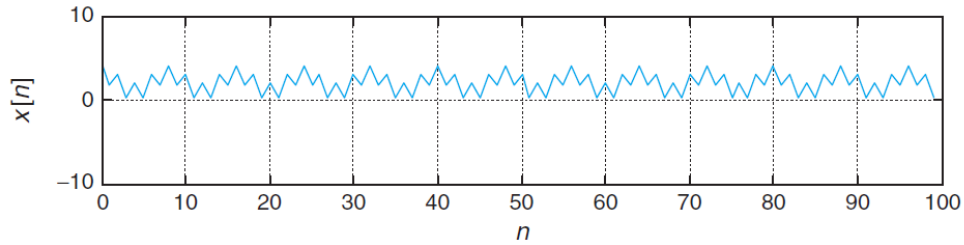
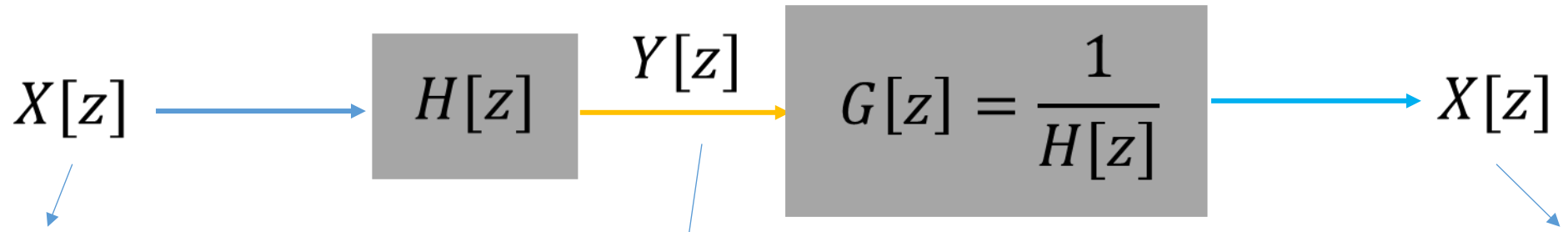
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$$G(z) = \frac{z^{11}}{z^{11} - 0.8}$$

This filter will remove the effect of multipath and give our original signal back.



Example signals.

Q. How can we use Z Transform to solve LCCDEs?

Next: Let's see how the Z Tx and its properties help solve LCCDEs

(recall that we previously solved these iteratively...)

$$x[n] = (2)^{-n}u[n]$$

$$y[-1] = 11/6, y[-2] = 37/36$$

$$y[n] - 5y[n-1] + 6y[n-2] = 3x[n-1] + 5x[n-2]$$

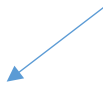
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$Y[z]$

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Z Tx has converted the difference equation into an algebraic equation!

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Using Transform Pairs
Table

$$y[n] = \left[\frac{26}{15} (0.5)^n - \frac{7}{3} (2)^n + \frac{18}{5} (3)^n \right] u[n]$$

**Let's see
another
example**

Zero-Input Response, Zero-State Response, and Total Response
Using Z Transform

Find the zero-input, zero-state, and total response of the DT-LTI system given by the difference equation

$$y[n] - \frac{1}{4}y[n-2] = x[n-1]$$

With input and initial conditions

$$\begin{aligned}x[n] &= 3u[n-5] \\ y[-2] &= y[-1] = 1\end{aligned}$$

We obtain **zero-state response** by solving the difference equation with initial conditions (ICs) set to zero.

Taking the z -transform of the difference equation (0 ICs) yields

$$Y_{\text{zsr}}[z] - \frac{1}{4}z^{-2}Y_{\text{zsr}}[z] = z^{-1}X[z].$$

Since $x[n] = 3u[n - 5]$, we see that

$$Y_{\text{zsr}}[z] \left(1 - \frac{1}{4}z^{-2}\right) = z^{-1}3z^{-5}\frac{z}{z-1}.$$

Solving for $Y_{\text{zsr}}[z]$ yields

$$Y_{\text{zsr}}[z] = \frac{3z^{-5}}{(z-1)(1-\frac{1}{2}z^{-1})(1+\frac{1}{2}z^{-1})} = \frac{3z^{-3}}{(z-1)(z-\frac{1}{2})(z+\frac{1}{2})}.$$

The modified partial fraction expansion is thus

$$\frac{Y_{\text{zsr}}[z]}{z} = 3z^{-4} \left(\frac{\frac{4}{3}}{z-1} + \frac{-2}{z-\frac{1}{2}} + \frac{\frac{2}{3}}{z+\frac{1}{2}} \right).$$

Inverting, we obtain

$$y_{\text{zsr}}[n] = \left(4 - 6\left(\frac{1}{2}\right)^{n-4} + 2\left(-\frac{1}{2}\right)^{n-4}\right) u[n - 4].$$

We find **zero-input response** by setting the input $x[n]$ to zero

$$Y_{\text{zir}}[z] - \frac{1}{4} [z^{-2}Y_{\text{zir}}[z] + z^{-2}(y[-1]z^1 + y[-2]z^2)] = 0.$$

$$Y_{\text{zir}}[z] (1 - \frac{1}{4}z^{-2}) = \frac{1}{4} + \frac{1}{4}z^{-1}.$$

Thus,

$$\frac{Y_{\text{zir}}[z]}{z} = \frac{\frac{1}{4}(z+1)}{z^2 - \frac{1}{4}} = \frac{\frac{3}{8}}{z - \frac{1}{2}} + \frac{-\frac{1}{8}}{z + \frac{1}{2}}.$$

Inverting yields

$$y_{\text{zir}}[n] = \left(\frac{3}{8} \left(\frac{1}{2} \right)^n - \frac{1}{8} \left(-\frac{1}{2} \right)^n \right) u[n].$$

Total response is simply sum of the two

$$y[n] = y_{\text{zir}}[n] + y_{\text{zsr}}[n]$$

Questions?? Thoughts??

